WASP-135b

R-0170

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-135_250625` · created 2026-07-16T17:37:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460851.943 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.05
Transit depth (Rp/Rs)^2	2.79 +/- 1.67 [%]
Orbital Inclination (inc)	81.39 +/- 2.94
Airmass coefficient 1 (a1)	1.86 +/- 0.62
Airmass coefficient 2 (a2)	-0.6 +/- 0.21
Scatter in the residuals of the lightcurve fit is	28.65 %
Best Comparison Star	None
Optimal Aperture	5.67
Optimal Annulus	8.97
Transit Duration (day)	0.053 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.05 vs 0.139 (deviation 0.3σ)
Duration fit vs published	0.053 d vs 0.0687 d (deviation 0.34σ)
Mid-time O–C	32.1 min
Depth SNR	1.8
Residual scatter	28.65 %

Light curve

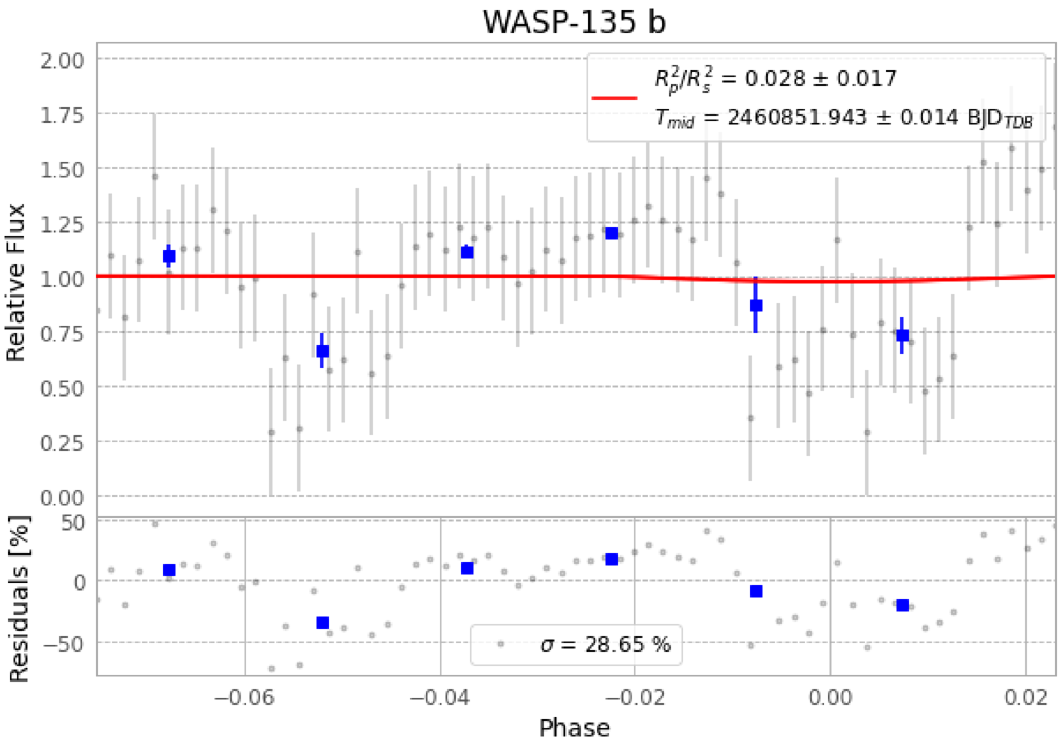

Figure 1 — FinalLightCurve_WASP-135 b_2025-06-25.png (SHA-256 5b2a7a916c8248ba…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 265], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 0a1c31f586423744…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62be8a4`

Data provenance

67 FITS frames (44 MB), `WASP-135250625080016.FITS` → `WASP-135250625111815.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250625.log (4116c982c299…), FinalLightCurve_WASP-135 b_2025-06-25.png (5b2a7a916c82…), FinalLightCurve_WASP-135 b_2025-06-25.csv (4a340dfe3efd…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

